

Advanced MERN Stack Developer Intern – Machine Test

Task Name: Project Management System (Advanced MERN Test)

Duration: 6–8 Hours

Difficulty: Intermediate–Advanced

Objective:

Evaluate candidate's ability to build a production-style MERN application with:

- Authentication (JWT)
- Role-based access (Admin & User)
- CRUD operations
- File upload
- Dashboard analytics
- Clean UI + API integration

Project Description:

Build a Project Management System where:

- Admins can manage users and projects.
- Users can view projects assigned to them and update progress.

1. Authentication & Authorization

Implement JWT-based login & signup.

User Roles:

- Admin
- User

Requirements:

- Password hashing using bcrypt.
- Only Admin can create Users.
- Admin can create/update/delete Projects.
- Users can only update project status.

2. Project Module (CRUD)

Fields:

- title (string)
- description (string)
- startDate (date)
- endDate (date)
- status (Pending / In-Progress / Completed)
- assignedUsers (array of user IDs)
- attachments (uploaded files)

3. File Upload

- Use Multer
- Max 3 attachments per project
- Local or Cloudinary storage

4. Dashboard & Analytics

API must return:

- Total users
- Total projects
- Count of projects by status
- Projects ending within 7 days

5. User Module

Admin:

- Create, update, delete users
- Change roles

User:

- Update own profile only

Frontend Requirements:

- Login Page
- Dashboard
- Project List
- Create Project
- Project Details
- Update Project Status
- User Management

Tech:

- Axios, Router
- Redux / Context API
- Clean UI (Material UI or Tailwind)

Submission Requirements:

- GitHub repo (frontend + backend)
- Postman Collection
- README
- (Bonus) Walkthrough video

Bonus Features:

- Theme toggle
- Real-time notifications (Socket.io)
- Activity logs
- Search, filter, pagination
- Deployment

Evaluation Criteria:

- Authentication & Security (20%)

- CRUD accuracy (20%)
- Code quality (20%)
- UI/UX (20%)
- File Upload + Dashboard (10%)
- Documentation & Git usage (10%)